Systems of Congruence

[Solve for x]

$12x^2 + 25x \equiv 10 \pmod{11}$

$12x^2 + 25x - 10 \equiv 0 \pmod{11}$

$x^2 + 3x - (-1) \equiv 0 \pmod{11}$

$x^2 + 3x + 1 \equiv 0 \pmod{11}$

$(x+a)(x+b) \equiv 0$

$x^2 + (a+b)x + ab \equiv 0$

$a + b \equiv 3 \pmod{11}$

$ab \equiv 1 \pmod{11}$

$a = 5,\ b = 9$ satisfies the system of equation

$(x+5)(x+9) \equiv 0 \pmod{11}$

$x \equiv 6$ or $2 \pmod{11}$

Chinese Remainder Theorem

[Solve for x]

$x \equiv 2 \pmod 3$ $\quad x \equiv 1 \pmod 4$ $\quad x \equiv 3 \pmod 5$

3, 4, 5 are pairwise coprime ✓ $\quad M = 3 \cdot 4 \cdot 5 = 60$

$m_1 = \frac{60}{3} = 20$ $\quad m_2 = \frac{60}{4} = 15$ $\quad m_3 = \frac{60}{5} = 12$

$20y_1 \equiv 1 \pmod 3$ $\quad 15y_2 \equiv 1 \pmod 4$ $\quad 12y_3 \equiv 1 \pmod 5$

$20 = 6 \cdot 3 + 2$ $\quad 15 = 3 \cdot 4 + 3$ $\quad 12 = 2 \cdot 5 + 2$

$3 = 1 \cdot 2 + \underline{1}$ $\quad 4 = 1 \cdot 3 + \underline{1}$ $\quad 5 = 2 \cdot 2 + \underline{1}$

$1 = 3 - 1 \cdot 2$ $\quad 1 = 4 - 1 \cdot 3$ $\quad 1 = 5 - 2 \cdot 2$

$= 3 - 1 \cdot (20 - 3 \cdot 6)$ $\quad = 4 - 1 \cdot (15 - 4 \cdot 3)$ $\quad = 5 - 2 \cdot (12 - 5 \cdot 2)$

$= 7 \cdot 3 - 1 \cdot 20$ $\quad = 4 \cdot 4 - 1 \cdot 15$ $\quad = -2 \cdot 12 + 5 \cdot 5$

$y_1 \equiv -1 \pmod 3$ $\quad y_2 \equiv -1 \pmod 4$ $\quad y_3 \equiv -2 \pmod 5$

$\equiv 2 \pmod 3$ $\quad y_2 \equiv 3 \pmod 4$ $\quad \equiv 3 \pmod 5$

$x \equiv \sum a m y \equiv 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 \equiv 233 \equiv 53 \pmod{60}$

[Solve $3^{302} \bmod 385$. Note $385 = 5 \cdot 7 \cdot 11$]

$3^{302} = (3^4)^{75} 3^2 \equiv 1 \cdot 3^2 \equiv 9 \equiv 4 \pmod 5$

$3^{302} = (3^6)^{50} 3^2 \equiv 1 \cdot 3^2 \equiv 9 \equiv 2 \pmod 7$

$3^{302} = (3^{10})^{30} 3^2 \equiv 1 \cdot 3^2 \equiv 9 \pmod{11}$

$x \equiv 4 \pmod 5$ $\quad x \equiv 2 \pmod 7$ $\quad x \equiv 9 \pmod{11}$

$m_1 = \frac{385}{5} = 77$ $\quad m_2 = \frac{385}{7} = 55$ $\quad m_3 = \frac{385}{11} = 35$

$77y_1 \equiv 1 \pmod 5$ $\quad 55y_2 \equiv 1 \pmod 7$ $\quad 35y_3 \equiv 1 \pmod{11}$

$\vdots$

$y_1 \equiv 3 \pmod 5$ $\quad y_2 \equiv 6 \pmod 7$ $\quad y_3 \equiv 6 \pmod{11}$

$x \equiv \sum a m y \equiv 4 \cdot 77 \cdot 3 + 2 \cdot 55 \cdot 6 + 9 \cdot 35 \cdot 6 \equiv 3474 \equiv 9 \pmod{385}$

$\therefore 3^{302} \equiv 9 \pmod{385}$

Fermat's Little Theorem

[Show 42 divides $n^7 - n$ for any n]

$42 \mid n^7 - n$ if $2, 3, 7 \mid n^7 - n$

$n^7 - n \equiv 0 \pmod 7$

$n^7 \equiv n \pmod 7$ ✓

$n^7 - n \equiv 0 \pmod 3$

$n^6 \cdot n \equiv n \pmod 3$

$(n^3)^2 n \equiv n \pmod 3$ $\Big\}$ by FLT, $n^3 \equiv n \pmod 3$

$n^2 n \equiv n \pmod 3$

$n^3 \equiv n \pmod 3$ ✓

$n^7 - n \equiv 0 \pmod 2$

$n^6 n \equiv n \pmod 2$

$(n^2)^3 n \equiv n \pmod 2$ $\Big\}$ by FLT

$n^3 n \equiv n \pmod 2$

$(n^2)^2 \equiv n \pmod 2$ $\Big\}$ by FLT

$n^2 \equiv n \pmod 2$ ✓

→ If n is composite, and $b^{n-1} \equiv 1 \pmod n$, n is called a $\boxed{\text{pseudoprime}}$ to base b.

→ If $b^{n-1} \equiv 1 \pmod n$, either prime or pseudoprime to base b.

→ If n is composite and $b^{n-1} \equiv 1 \pmod n$ for all b where $\gcd(b,n) = 1$, n is called a $\boxed{\text{Carmichael number}}$.

Let b be any positive integer s.t. $\gcd(b,n) = 1$.

Since p_j are distinct primes, $\gcd(b, p_j) = 1$ for any $j = 1, 2, \dots, k$.

If $p_j - 1 \mid n - 1$, there exists m_j s.t. $n - 1 = m_j(p - 1)$

$b^{n-1} = b^{m_j(p_j-1)} = \underbrace{(b^{p_j-1})^{m_j}}_{\text{by Fermat's Little Theorem}} \equiv 1^{m_j} \equiv 1 \pmod{p_j}$ for all j

$b^{n-1} = 1 \pmod n$ by Chinese Remainder Theorem

$2821 = 7 \times 403 = 7 \times 13 \times 31$

$n - 1 = 2820$ $\quad p_j = 6, 12, 30$

$6 \mid 2820$ ✓

$12 \mid 2820$ ✓

$30 \mid 2820$ ✓

Since 2821 is composite, can be factorized into distinct primes and for each prime divisor p_j, $p_j - 1 \mid 2821 - 1$

∴ 2821 is a Carmichael number

32. Send BUY NOW. $(n_A, e_A) = (2867, 7) = (61 \cdot 47, 7)$; $d_A = 1183$

$(n_B, e_B) = (3127, 21) = (59 \cdot 53, 21)$, $d_B = 1149$

BUY NOW → M : [1, 20, 24, 26, 13, 14, 22]

Signing: $\boxed{S \equiv M^{d_{Alice}} \bmod n_{Alice}} = m^{1183} \bmod 2867$

S : [1, 1785, 1322, 1638, 1178, 122, 1468]

Encrypt Signed Message: $\boxed{C \equiv S^{e_{Bob}} \bmod n_{Bob}} = s^{21} \bmod 3127$

C : [1, 1639, 2874, 1351, 1714, 1897, 1654]

1. $\boxed{\text{Pollard's Factoring Algorithm}}$

$f(x) = x^2 + 1 \pmod{7081}$ $\quad x_0 = 2$ $\quad y_0 = 2$

If $1 < \gcd(|x_i - y_i|, n) < n$, then we found a non-trivial factor

→ $x_1 = f(2) = 5 \pmod{7081}$ $\quad$ → $x_2 = f(5) = 26 \pmod{7081}$

$y_1 = f(f(2)) = f(5) = 26 \pmod{7081}$ $\quad y_2 = f(f(26)) = f(677) = 5146 \pmod{7081}$

$|x_1 - y_1| = 21$ $\quad |x_1 - y_1| = 5120$

$\gcd(21, 7081) = 1$ $\quad \gcd(5120, 7081) = 1$

→ $x_3 = f(26) = 677$

$y_3 = f(f(5146)) = f(7079) =$

$|x_3 - y_3| = 6402$

$\gcd(6402, 7081) = 97$

Since the gcd $\neq 1$, 97 is a non-trivial prime factor.

⇒ $7081 = 97 \times 73$

Cryptography

24. Encrypt ATTACK with $n = 43 \cdot 59$ and $e = 13$

A = 0, B = 1, C = 2 … Z = 25

ATTACK → [0, 19, 19, 0, 2, 10]

$\boxed{\text{RSA encryption}}$

$\boxed{c = m^e \bmod n}$

(0, 19) → 0190 $\quad$ (19, 0) → 1900 $\quad$ (2, 10) → 0210

$190^{13} \bmod 2537 = 1448$ $\quad 1900^{13} \bmod 2537 = 1867$ $\quad 210^{13} \bmod 2537 = 2117$

26. Decrypt 3185 2038 2460 2550 with $n = 53 \cdot 61$ and $e = 17$

$$\boxed{\begin{aligned} \phi n &= (p-1)(q-1) = 52 \cdot 60 = 3120 \\ d \cdot e &\equiv 1 \pmod{\phi n} \end{aligned}}$$

$d \cdot 253 \equiv 1 \pmod{3120}$

$3120 = 12 \cdot 253 + 84$

$253 = 3 \cdot 84 + \underline{1}$

$1 = 253 - 3 \cdot 84$

$= 253 - 3(3120 - 12 \cdot 253)$

$= \underline{37} \cdot 253 - 3 \cdot 3120$

$d \equiv 37 \pmod{3120}$

$\boxed{m = c^d \bmod n}$ $\boxed{\text{RSA decryption}}$

$2642 \bmod 3233 = 614$

$3218^{37} \bmod 3233 = 2601$

$2887^{37} \bmod 3233 = 400$

$1068^{37} \bmod 3233 = 1718$

go
_b
ea
rs

go bears

28. Prove this. $\boxed{C \equiv M^e \pmod n \ \text{(encryption)} \quad C^d \equiv M \pmod n \ \text{(decryption)}}$

$C \equiv M^e \pmod p$ $\quad C \equiv M^e \pmod q$

$C^d \equiv (M^e)^d \pmod p$ $\quad C^d \equiv (M^e)^d \pmod q$

By definition of inverses, $e \cdot d \equiv 1$ $\quad$ By definition of inverses, $e \cdot d \equiv 1$

$C^d \equiv M \pmod p$ $\quad C^d \equiv M \pmod q$

From CRT, since p and q are coprime, there exists a unique solution $C^d \equiv M \pmod n$ such that $n = pq$.

30. Generate shared key using $p = 101$ and $a = 2$. Alice selects $k_1 = 7$, Bob selects $k_2 = 9$ $\boxed{\text{Diffie-Hellman Key}}$

$p = 101$, $a = 2$, $k_1 = 7$, $k_2 = 9$

Alice's public key $A = a^{k_1} \bmod p = 2^7 \bmod 101$

$= 27$ → sends to Bob

Bob's public key $B = a^{k_2} \bmod p = 2^9 \bmod 101$

$= 7$ → sends to Alice

Alice's shared secret key $K = B^{k_1} \bmod p = 7^7 \bmod 101$

$= 90$

Bob's shared secret key $K = A^{k_2} \bmod p = 27^9 \bmod 101$

$= 90$

2a) $\boxed{\text{Fermat's Factoring Algorithm}}$

$a^2 \equiv b^2 \pmod n$

$a^2 - b^2 \equiv 0 \pmod n$

$(a+b)(a-b) = 0 \pmod n$

Assume $a \not\equiv b \pmod n$ and $a \not\equiv -b \pmod n$.

Therefore, $(a+b) \not\equiv 0 \pmod n$ and $(a-b) \not\equiv 0 \pmod n$

So, $\gcd(a+b, n) \neq n$ because $a+b$ is not divisible by n and $\gcd(a-b, n) \neq n$ because $a-b$ is not divisible by n

Also $\gcd(a+b, n) = 1$ or $\gcd(a-b, n) = 1$ cannot be true, since it will mean either term and n are coprime, but contradicts the fact that they share a factor n since $a^2 \equiv b^2 \pmod n$.

2b) $\sqrt{6893} \approx 83$

$a = 84, 85, 86 \dots$ to find $(a^2 \bmod n)$ be a square

$84^2 \bmod 6893 = 163$ ✗

$85^2 \bmod 6893 = 332$ ✗

$86^2 \bmod 6893 = 503$ ✗

$87^2 \bmod 6893 = 676$ ✓ $\quad 26^2 = 676$

$a = 87$ and $b = 26$

$\gcd(a+b, n) = \gcd(113, 6893) = 113$ ✓

$\gcd(a-b, n) = \gcd(61, 6893) = 61$ ✓

⇒ $6893 = 113 \times 61$

Induction

22. For which non-negative integer n is $n^2 \le n!$
Prove your answer.

a) Base case: $n = 4$
$4^2 \le 4!$
$16 \le 24$ ✓

$0^2 \le 0!$ ✓
$1^2 \le 1!$ ✓
$2^2 \le 2!$ ✗
$3^2 \le 3!$ ✗
$4^2 \le 4!$ ✓
$5^2 \le 5!$ ✓
⋮

b) Assume $k^2 \le k!$

c)
$$(k+1)^2 \le (k+1)!$$
$$k^2+2k+1 \le k!(k+1)$$
$$k^2+2k+1 \le k^2(k+1)$$
$$k^2+2k+1 \le k^3+k^2$$
$$2k+1 \le k^3$$

For $n = 0, 1$ and $n \ge 4$, $n^2 \le n!$ holds true.

76. Suppose we want to prove $\frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2n-1}{2n} < \frac{1}{\sqrt{3n}}$ for all positive integers n.

(a) Show if we try and use induction, base case works but inductive case fails.

a) Base case: $n = 1$
$$\frac{2(1)-1}{2(1)} < \frac{1}{\sqrt{3}}$$
$$\frac{1}{2} < \frac{1}{\sqrt{3}} \checkmark$$

b) Assume $\frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2k-1}{2k} < \frac{1}{\sqrt{3k}}$

c) $\frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2k-1}{2k}\cdot\frac{2(k+1)-1}{2(k+1)}$
$$\frac{1}{\sqrt{3k}}\cdot\frac{2k+1}{2k+2} \le \frac{1}{\sqrt{3(k+1)}}$$
$$\frac{1}{3k}\cdot\frac{(2k+1)^2}{(2k+2)^2} \le \frac{1}{3(k+1)}$$
$$(2k+1)^2(k+1) \le k(2k+2)^2$$
$$(4k^2+4k+1)(k+1) \le k(4k^2+8k+4)$$
$$\cancel{4k^3}+\cancel{4k^2}+k+\cancel{4k^2}+4k+1 \le \cancel{4k^3}+8k^2+4k$$
$$1 \le -k$$
$$0 \le -(k+1) \quad \text{Contradiction}$$

(b)(a) Base case: $n = 2$
$$\frac{1}{2}\cdot\frac{3}{4} < \frac{1}{\sqrt{6}}$$
$$\frac{3}{8} < \frac{1}{\sqrt{6}}$$
$$\frac{3}{8} < \frac{3}{7.34\ldots} \checkmark$$

(b) Assume $\frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2k-1}{2k} < \frac{1}{\sqrt{3k+1}}$

(c) $\frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2k-1}{2k}\cdot\frac{2(k+1)-1}{2(k+1)}$
$$\frac{1}{\sqrt{3k+1}}\cdot\frac{2(k+1)-1}{2(k+1)} \le \frac{1}{\sqrt{3(k+1)+1}}$$
$$\frac{1}{\sqrt{3k+1}}\cdot\frac{2k+1}{2k+2} \le \frac{1}{\sqrt{3k+4}}$$
$$\frac{1}{3k+1}\cdot\frac{(2k+1)^2}{(2k+2)^2} \le \frac{1}{3k+4}$$
$$(2k+1)^2(3k+4) \le (2k+2)^2(3k+1)$$
$$0 \le k \quad \checkmark \text{ for all } k \ge 1$$

62. Show that n lines separate the plane into $\frac{n^2+n+2}{2}$ regions if no two lines are parallel and no three lines pass thru a common point.

a) Base case: $n = 1$ [1 | 2] $\frac{1^2+1+2}{2} = \frac{4}{2} = 2$ regions ✓

b) Assume k lines separate plane into $\frac{k^2+k+2}{2}$ regions if no two lines are parallel and no three lines pass thru a common point.

c) Adding the $(k+1)^{th}$ line, because no two lines are parallel, this new $(k+1)^{th}$ line meets each of the k existing line at k distinct pts. The pts must be unique since no three lines pass through the same pt. Thus the new line has $(k+1)$ segments as a result of those k intersection pts.

Each of the segments pass through an existing region and split it in half, effectively increasing the # of regions by $(k+1)$.

[1 | 2] [1 | 2 / 3 | 4] [1 2 3 4 5 6 7]

$$\text{\# of region with } k \text{ lines} + (k+1) = \frac{k^2+k+2}{2} + k+1 = \frac{k^2+k+2+2k+2}{2} = \frac{k^2+3k+4}{2}$$
$$\text{\# of region with } (k+1) \text{ lines} = \frac{(k+1)^2+(k+1)+2}{2} = \frac{k^2+2k+1+k+1+2}{2} = \frac{k^2+3k+4}{2}$$

8. Suppose a store offers gift certificates in denominations of 25/40 dollars. Find possible total amounts you can form with these certificates.

$n = 25x + 40y$

(a) Base cases:

For $n = 0$: $0 = 25(0) + 40(0)$ ✓
For $n = 25$: $25 = 25(1) + 40(0)$ ✓
For $n = 40$: $40 = 25(0) + 40(1)$ ✓
For $n = 50$: $50 = 25(2) + 40(0)$ ✓
For $n = 65$: $65 = 25(1) + 40(1)$ ✓
For $n = 80$: $80 = 25(0) + 40(2)$ ✓

(b) Assume for $k \ge 80$, we can find positive integers x, y such that $k = 25x + 40y$

(c)
$$25(x+1) + 40y = k+25$$
$$25x + 40(y+1) = k+40$$
Since we can form $k+25$ and $k+40$ assuming we can form amounts from 80 to k, there exists positive integers x that $n - 25x + 40y$.

10. $n-1$ breaks needed to break to n separate squares. Since each break increases the number of squares by one.

(a) Base case: break to $n = 1$ separate squares
$1 - 1 = 0$ no breaks required ✓

(b) Assume for $k \ge 1$, breaking to k separate squares require $k-1$ breaks.

(c) For a bar with $k+1$ squares, breaking it requires one additional break.
$$\underset{\text{first break}}{1} + \underbrace{(k-1)}_{\text{by assumption}}^{\text{\# of breaks for remaining } k \text{ squares}} = k \rightarrow \text{matches formula when subbing } k+1: (k+1)-1 = k \checkmark$$

12. f_n is the n-th Fibonacci number. Prove that $f_1^2 + f_2^2 + \cdots + f_n^2 = f_n f_{n+1}$ when n is a positive integer.

$f_1 = 1 \quad f_2 = 1 \quad f_k = f_{k-1} + f_{k-2}$ for $k \ge 3$

(a) Base case: $n = 1$
$$f_1^2 = f_1 f_2$$
$$1 = 1 \cdot 1 \checkmark$$

(b) Assume $f_1^2 + f_2^2 + \cdots + f_k^2 = f_k f_{k+1}$ for any k.

(c)
$$f_1^2 + f_2^2 + \cdots + f_k^2 + f_{k+1}^2$$
$$= f_k f_{k+1} + f_{k+1}^2$$
$$= f_{k+1}(f + f_{k+1})$$
$$= f_{k+1} f_{k+2} \quad \text{true when we substitute } k = k+1$$

2(a) $f^{(0)}(x) = x \quad f^{(1)}(x) = x^2 \quad f^{(2)}(x) = f(f(x)) = x^4$
$f^{(3)}(x) = x^8 \quad \ldots \quad f^{(n)}(x) = x^{2^n}$

(a) Base case $n = 0$
$$f^{(0)}(x) = x^{2^0} = x^1 = x$$

(b) Assume $f^{(k)}(x) = x^{2^k}$ for some $k \ge 0$.

(c)
$$f^{(k+1)}(x) = f(f^{(k)}(x))$$
$$= f(x^{2^k}) \quad \text{by inductive hypothesis}$$
$$= (x^{2^k})^2$$
$$= x^{2\cdot 2^k}$$
$$= x^{2^{k+1}} \checkmark \quad \text{true when subbing } k = k+1$$

(b) (a) Base case: $m = 0$ (induction on m)
$$f^{(0+n)}(x) = f^{(n)}(x) \checkmark$$
$$f^{(0)}(f^{(n)}(x)) = f^{(n)}(x) \checkmark$$

(b) Assume $f^{(k+n)}(x) = f^{(k)} f^{(n)}(x)$ for some $k \ge 0$ and all $n \ge 0$

(c)
$$f^{((k+1)+n)}(x) = f^{(k+n+1)}(x)$$
$$= f(f^{(k+n)}(x)) \quad \text{by definition of } f$$
$$= f(f^{(k)}(f^{(n)}(x))) \quad \text{by inductive hypothesis}$$
$$= f^{(k+1)}(f^{(n)}(x)) \checkmark \quad \text{true when subbing } k = k+1$$

Binomial Coefficients

$$(x+y)^n = \sum_{j=0}^{n}\binom{n}{j}x^{n-j}y^j \qquad \binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

$$\sum_{k=0}^{n}\binom{n}{k} = 2^n \qquad \sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0 \qquad \sum_{k=0}^{n}2^k\binom{n}{k} = 3^n$$

$P(n,r) = \frac{n!}{(n-r)!}$ order matters $\quad C(n,r) = \frac{n!}{r!(n-r)!}$ order doesn't matter

12a) x^6y^9's coefficient in $(5x^2 + 2y^3)^6$
$$\binom{6}{k}(5x^2)^{6-k}(2y^3)^k$$
$12 - 2k = 6 \quad 3k = 9$
$k = 3 \quad k = 3$ ✓

Coefficient: $\binom{6}{3}\cdot 5^{6-3}\cdot 2^3 = 20000$

e) x^8y^9's coefficient
$12 - 2k = 8 \quad 3k = 9$
$k = 2 \quad k = 3$
Coefficient: 0

r-permutations of a set of n objects: n^r
$C(n+r-1, r) = C(n+r-1, n-1)$ r-combinations of a set of n objects
WITH REPETITIONS

$\frac{n!}{n_1!n_2!\cdots n_k!}$ permutations of n objects with indistinguishable objects

10 a) We have five flavors. Ways to choose a dozen.
$$\binom{\overset{\text{\# of types}}{6}+\overset{\text{\# of items}}{12}-1}{12} = \binom{17}{5} = \frac{17!}{12!\,5!} = 6188 \text{ ways}$$

c) Two dozen with at least two of each kind
2 croissants per type: 2×6 croissants
$24 - 12 = 12$ croissants to distribute freely
$$\binom{6+12-1}{12} = 6188 \text{ ways}$$

d) Two dozen with no more than two broccoli flavor.
$$\underset{0 \text{ broccoli}}{\binom{5+24-1}{24}} + \underset{1 \text{ broccoli}}{\binom{5+23-1}{23}} + \underset{2 \text{ broccoli}}{\binom{5+22-1}{22}}$$
$= 20475 + 17550 + 14950$
$= 52975$ ways

20. # of solution to the inequality
$$x_1 + x_2 + x_3 \le 11$$
$$x_1 + x_2 + x_3 + x_4 = 11 \quad \text{convert inequality to equality}$$
$$\binom{\overset{\text{\# of variables}}{4}+11-1}{\underset{\text{sum we want}}{11}} = \binom{14}{11} = 364 \text{ ways}$$

22. ways to be scheduled for 5 days in October with consecutive days
$$\binom{6+22-1}{22} = 80730 \text{ ways}$$

28. # of pos. integers < 1,000,000 & 1 digit = 9 & sum of digits = 13
$$\binom{5+4-1}{4} = \binom{8}{4} = 70 \text{ ways}$$
Choose 1 position out of 6 for the digit 9
Remaining 5 digits must sum to 4.
$6 \times 70 = 420$ ways for all digits

40. a) $\frac{40!}{10!\,10!\,10!\,10!} = \frac{40!}{(10!)^4}$ Distinguishable

b) $\frac{1}{4!}\cdot\frac{40!}{10!\,10!\,10!\,10!} = \frac{40!}{24\cdot(10!)^4}$ Indistinguishable